

Felipe Yanez Leaño

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Biomedical Engineer | Data Scientist | Machine Learning & AI for Healthcare

Professional Profile Data Analyst/Scientist with experience in health care research, data analytics and applications of machine learning and artificial intelligence as well. Utilizing sophisticated analytical workflows in data acquisition, preprocessing, statistical analysis, predictive modeling and data visualization for evidence-based decision-making.

Implemented data-driven solution through Python, SQL, machine learning, computer vision and statistical methods in research / healthcare settings. Demonstrated experience in working with cross-functional teams, presenting complex analytical insights to both technical and non-technical audience, translating data into actionable insights. A highly motivated and demonstrated SME in healthcare utilization management/HL7 implementation with a strong background in using AI, advanced analytics and clinical data to innovate the care delivery process.

Technical Skills

Python | SQL | Pandas | NumPy | Scikit-learn | Machine Learning | Artificial Intelligence | Statistical Analysis | Predictive Modeling | Computer Vision | Signal Processing | Data Analytics | Data Visualization | Exploratory Data Analysis (EDA) | Data Modeling | Workflow Automation | Clinical & Biomedical Data Analysis | Cross-functional Collaboration

Professional Experience

Biomedical Research Engineer – Data Analysis & Machine Learning

Fundación Clínica Shaio | Bogotá, Colombia – Research Group EXPLORATORIUM

August 2021 – Present

- Created Python-based analytic pipelines for biomedical signal processing, clinical data analysis, and computer vision applications in healthcare research.
- Developed, trained and assessed machine learning and artificial intelligence models for discriminating physiological/specific biological variables from multimodal (i.e., video + sensor data).
- Managed technical activities organized data analysis workflows for research and innovation/new technology-development projects by collaborating with multidisciplinary teams.
- Statistical analysis, model validation and data interpretation supporting evidence-based research and healthcare technology development

Highlights

- Principal Investigator for multiple AI and Computer Vision projects that include video-based physiological monitoring, hand hygiene detection, remote heart-rate estimation.
- Designed, developed and validated a polyphonic controller based EEG neurofeedback solution integrated with biomedical sensing, embedded system, data acquisition and analytical processing (4-channel).

Adjunct Lecturer in Bioinstrumentation

Pontificia Universidad Javeriana | Bogotá, Colombia

January 2026 – Present

- Teach Bioinstrumentation courses to undergraduate students in Bioengineering and Electronic Engineering programs.
- Design and deliver theoretical and practical sessions covering biomedical instrumentation, physiological signal acquisition, sensors, measurement systems, and data analysis principles.
- Guide students in applying engineering methodologies to solve challenges in healthcare technology and biomedical systems.
- Mentor students in laboratory activities, project development, and technical problem-solving.

Highlights

- Responsible for the instruction and assessment of undergraduate students in multidisciplinary engineering programs.
- Integrate concepts from bioengineering, instrumentation, signal processing, and data analytics into

course activities and laboratory sessions.

- Currently teaching a cohort of 18 undergraduate students.

Adjunct Lecturer in Bioinstrumentation

Escuela de Ingeniería Julio Garavito | Bogotá, Colombia

August 2021 – July 2022

- Taught theoretical and practical bioinstrumentation to undergraduate students.
- Designed and supervised laboratory sessions, ensuring safe and effective hands-on learning.
- Mentored students in applying bioinstrumentation tools and measurement techniques.

Highlights

- Responsible instructor for two groups of 24+ students.
- Promoted to Instructor rank based on performance and experience.

Biomedical Research Engineer

Research Assistant for Dr. Jorge Reynolds Pombo | Bogotá, Colombia

January 2020 – August 2021

- Developed Python-based data processing and analytical workflows for cardiac and neurological signal analysis, supporting biomedical research and clinical data interpretation.
- Applied statistical methods, signal processing techniques, and data modeling approaches to improve the analysis and characterization of physiological data.
- Led and authored research projects in cardiovascular medicine, transforming complex biomedical datasets into actionable research findings and clinical insights.
- Collaborated with multidisciplinary teams of researchers, clinicians, and engineers to support healthcare innovation and scientific dissemination.
- Coordinated the planning and execution of medical technology events and scientific symposia in partnership with Fundación Clínica Shaio, promoting knowledge transfer and research visibility.

Data Science Projects

Film Junky IMDb Review Classification – TripleTen Bootcamp | 2025

Sentiment classification of IMDb reviews using Python, Pandas, Scikit-learn (DummyClassifier and supervised models), and Matplotlib.

Achievement: Reached 80% accuracy, validating machine learning approaches for text data.

GitHub: https://github.com/cfelipe-yanez/film_junky_imdb_review_classification

Sweet Lift Taxi Demand Prediction – TripleTen Bootcamp | 2025

Hourly taxi demand prediction using Python, Scikit-learn, Matplotlib, Seaborn, and time series analysis.

Achievement: Developed optimized models with competitive RMSE performance, improving driver scheduling and planning.

GitHub: https://github.com/cfelipe-yanez/sweet_lift_taxi_demand_prediction

Beta Bank Customer Churn Prediction – TripleTen Bootcamp | 2025

Prediction of bank customer churn using Python, Scikit-learn, Pandas, Seaborn, and a complete data pipeline.

Achievement: Optimized F1 score and AUC-ROC, supporting customer retention strategies.

GitHub: <https://github.com/cfelipe-yanez/beta-bank-customer-churn-prediction>

Education

Data Scientist Bootcamp, TripleTen

2025

Biomedical Engineer, Escuela Colombiana de Ingeniería Julio Garavito / Universidad del Rosario

2019

High School Diploma, Colegio San Viator, Bogotá

2013

Languages:

- **English**, Upper-Intermediate (B2) | Certification: IELTS
- **Spanish**, Native
- **Portuguese**, Basic (A2)

June 2019